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GEOPHYSICS

Cascading magmatic unrest and aseismic faulting in the Aegean Sea enabled by elastic stress relay

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Magma reservoirs commonly exist in clusters and situate within complex fault networks, yet sequential triggering of unrest at multiple reservoirs and faults is rarely observed. We show that the 2025 earthquake swarm in the Christiana-Santorini-Kolumbo complex of the Aegean Sea was driven by elastic stress transfer between two magma reservoirs, a 9-kilometer-long dike, and a 17-kilometer-long, southeast-dipping fault. Pressurization of the magma reservoir under Santorini triggered a dike intrusion originating from a separate magma reservoir under the Kolumbo volcano. The dike, in turn, triggered a moment magnitude (M_w) 6.2 aseismic slip transient on a SW-NE trending branch of the Amorgos fault. Episodic bursts of upward-propagating seismicity within the geodetically constrained dike potentially indicate buoyancy-driven, segmented dike opening. We suggest that unrest in rift zones can propagate over spatial scales much larger than individual magma reservoirs and faults via sequential relay of elastic stress.

INTRODUCTION

Numerous volcanic eruptions (*1*–*7*) and earthquakes (*8*–*13*) are thought to be triggered by stress transfer between magmatic systems and crustal faults. These stress perturbations, whether due to static deformation or seismic waves, are commonly small compared with the shear strength of a fault or the tensile strength of the crust surrounding a magma reservoir, suggesting that tectonically active regions are often critically stressed. Magma reservoirs tend to exist in clusters (*14*–*18*) and volcanoes tend to situate within extensive fault systems (*19*–*21*). In these complex settings, multiple critically stressed magma reservoirs and faults may interact with one another via stress perturbations. However, observations of triggered unrest rarely involve more than a single stress perturbation source-receiver pair. The paucity of observations of complex triggering sequences may result from insufficient monitoring, because, globally, most volcanic regions lack advanced monitoring systems (*22*). Nonetheless, stress triggering among multiple faults and magmatic systems would substantially compound hazard potentials and needs to be understood.

The Christiana-Santorini-Kolumbo (CSK) volcanic complex, a continental rift zone in the Aegean Sea, is a natural laboratory for studying volcanotectonic interactions and associated hazards. The CSK complex (Fig. 1A) comprises a linear trend of volcanoes embedded in a system of SW-NE trending extensional basins and faults (*23*, *24*). The major volcanoes within the complex are the subaerial Santorini volcano and the submarine Kolumbo volcano (Fig. 1A). While shallow (<5 km in depth) magma reservoirs have been imaged immediately beneath Santorini (*25*) and Kolumbo (*26*), low-velocity anomalies in the crust also indicate the presence of mush and magma storage regions throughout the CSK that together occupy a greater lateral extent than the footprint of either individual volcano (*25*, *27*, *28*). The volcanoes and faults in the region present major eruptive and seismic hazards: the Amorgos fault produced the 1956 moment magnitude (M_w) 7.8 Amorgos earthquake, the largest earthquake in Greece in the 20th century (*29*); and the Santorini volcano hosted large (VEI ~ 7) caldera-forming explosive eruptions recurring every 10,000 to 30,000 years (*30*).

In late January 2025, a productive earthquake swarm began near the Kolumbo volcano, 15 km to the northeast of Santorini (*31*). During the initial phase of the swarm (26 January to 20 February), the local seismic network reported hundreds of earthquakes per day, with magnitudes up to M_w 5.2 (*32*). Seismic activity began slowing in mid-February. As of July 2025, earthquake rates in the region have not yet returned to background levels.

The complex volcanotectonic setting of the CSK complex has contributed to conflicting interpretations of the processes driving the 2025 sequence. Analyses of seismic and geodetic data have led workers to argue variously for dike-driven unrest (*31*, *33*), diking with concurrent dike-driven conjugate faulting (*34*), and aseismic slip-dominated deformation (*35*). Furthermore, sequential inflation at Santorini and deflation at Kolumbo hints at coupling between the two volcanoes' magma reservoirs, but the mechanisms underlying this coupling (e.g., stress transfer or hydraulic connection) are uncertain (*33*). These interpretations are pertinent to our understanding of both volcanic hazard in the CSK and, more generally, the mechanisms of volcanotectonic interactions in rift systems. To address these questions, we construct an enhanced seismic catalog capturing the swarm with a machine learning–based workflow and analyze deformation measured by Global Navigation Satellite System (GNSS) and Interferometric Synthetic Aperture Radar (InSAR). Our catalog shows that the seismic swarm was initiated by a dike intrusion from a reservoir below Kolumbo. Our InSAR deformation measurements on the islands of Amorgos, Anydros, Ios, Anafi, and Santorini, in conjunction with previously reported GNSS deformation measurements (*35*), indicate that a M_w 6.2 aseismic slip transient on a branch of the Amorgos fault occurred concurrently with the dike intrusion.

RESULTS

The seismic swarm was preceded by 7 months of gradual, inflationary deformation centered on Nea Kameni (Fig. 1A, inset) and accompanied by 1 month of rapid, deflationary deformation concentrated on the northeastern corner of Thera (Fig. 1, A and B). During the swarm, Ios and Anafi symmetrically tilted away from the SW-NE axis of the

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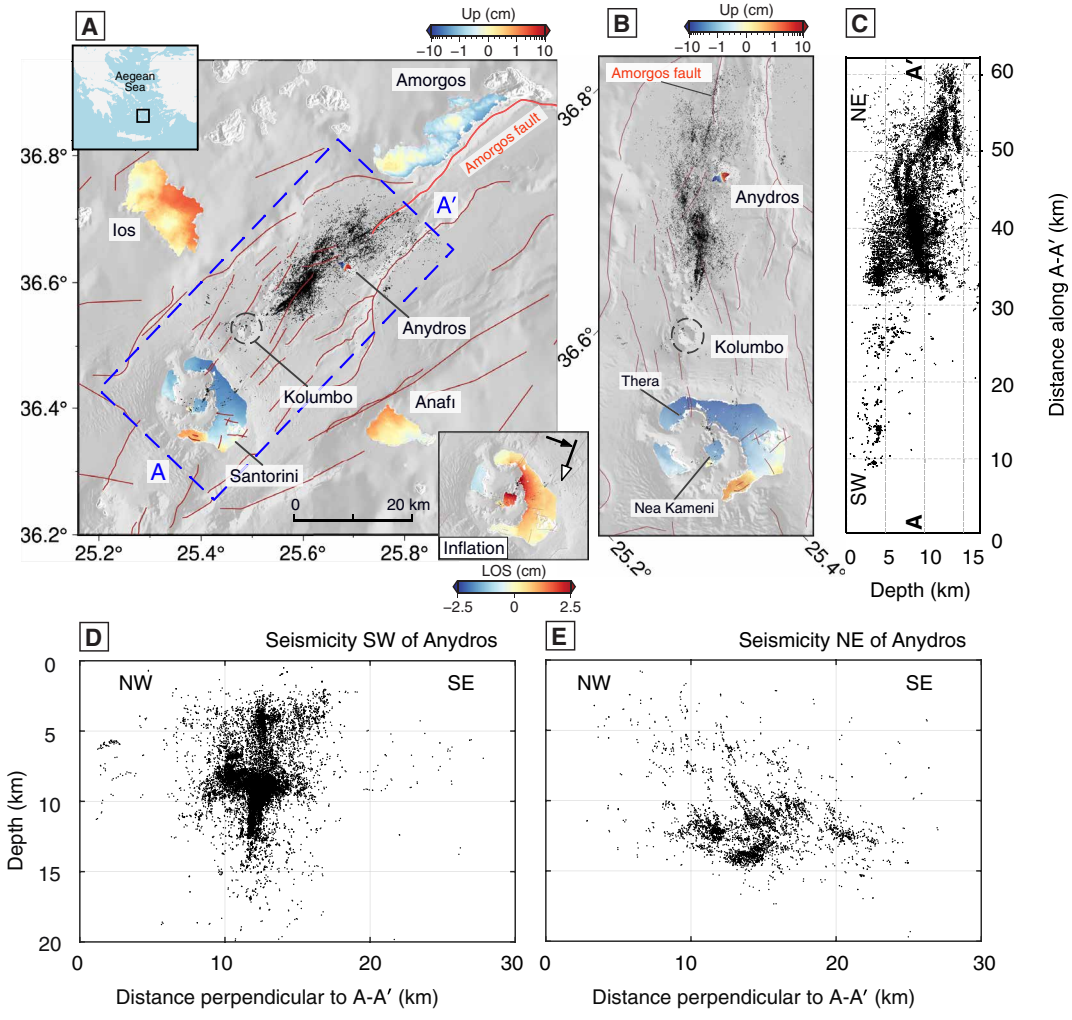


Fig. 1. Seismicity and deformation in the CSK volcanic complex in the Aegean Sea spanning the 2025 volcanotectonic sequence. (**A**) InSAR-derived vertical deformation (cumulative from 27 January to 29 June 2025; see table S6) plotted in color; black dots are relocated seismicity (from 8 January 2024 to 12 April 2025); brown lines are mapped faults in the National Observatory of Athens (NOA) fault database (*72*). Lower-right inset of (A): Line-of-sight (LOS) inflationary deformation on Santorini (for the descending track; cumulative from 1 July 2024 to 26 January 2025). (**B**) Seismicity and deformation along the A-A′ profile boxed in (A). (**C**) Depth section of seismicity along A-A′. Seismicity in the vicinity of Santorini and Kolumbo is plotted with a larger marker size to increase visibility. (**D** and **E**) Seismicity southwest of Anydros (defined as <45 km along A-A′) and northeast of Anydros (defined as >50 km along A-A′) shown in a depth section perpendicular to A-A′.

CSK complex, while Anydros tilted toward the northwest (Fig. 1, A and B). The swarm extended for more than 50 km along the CSK complex, with earthquakes deepening from 0 to 5 km near Santorini, in the southwest, to 5 to 15 km near Anydros, in the northeast (Fig. 1, B and C). Hypocenters between Santorini and Anydros delineate a subvertical plane striking NE (Fig. 1D), whereas hypocenters between Anydros and Amorgos concentrate along multiple curved planes dipping to the southeast (Fig. 1E).

From these observations, we deduce that underlying the seismic crisis of 2025 is a cascading sequence of events involving (i) the pressurization of a magma reservoir beneath Santorini over a 7-month period; (ii) depressurization of a magma reservoir beneath Kolumbo volcano, accompanied by (iii) an ~8.7-km-long dike intrusion along the SW-NE trending CSK complex; and (iv) a M_w 6.2 oblique, dip-slip aseismic transient on a SW-NE trending branch of the Amorgos fault, over a 1-month period.

Preswarm unrest at Santorini volcano

Before July 2024, seismicity in our catalog was sparse (Fig. 2A), and there was little deformation at Santorini (Fig. 2C). Beginning in July 2024, seismic activity accelerated beneath Santorini to a roughly constant rate of 1.2 events/day (Fig. 2A). The increase in seismicity coincided with the onset of inflation at Santorini; there was no detectable deformation at Anydros during this period (Fig. 2C). In August 2024, seismicity near Kolumbo volcano accelerated to 0.6 events/day (Fig. 2A). The hypocenters of most of these events have only unrelocated solutions, because they are too small and spatially scattered to produce useful cross-correlation values for relative relocation (especially given the poor station coverage; figs. S1 and S2).

The inflationary deformation at Santorini (Fig. 2C) spanned a period of ~7 months (from 1 July 2024 to 26 January 2025) and can be explained by a volume increase of 1.1×10^{-2} to 1.3×10^{-2} km^3 (90% confidence interval) in a spheroidal magma reservoir located

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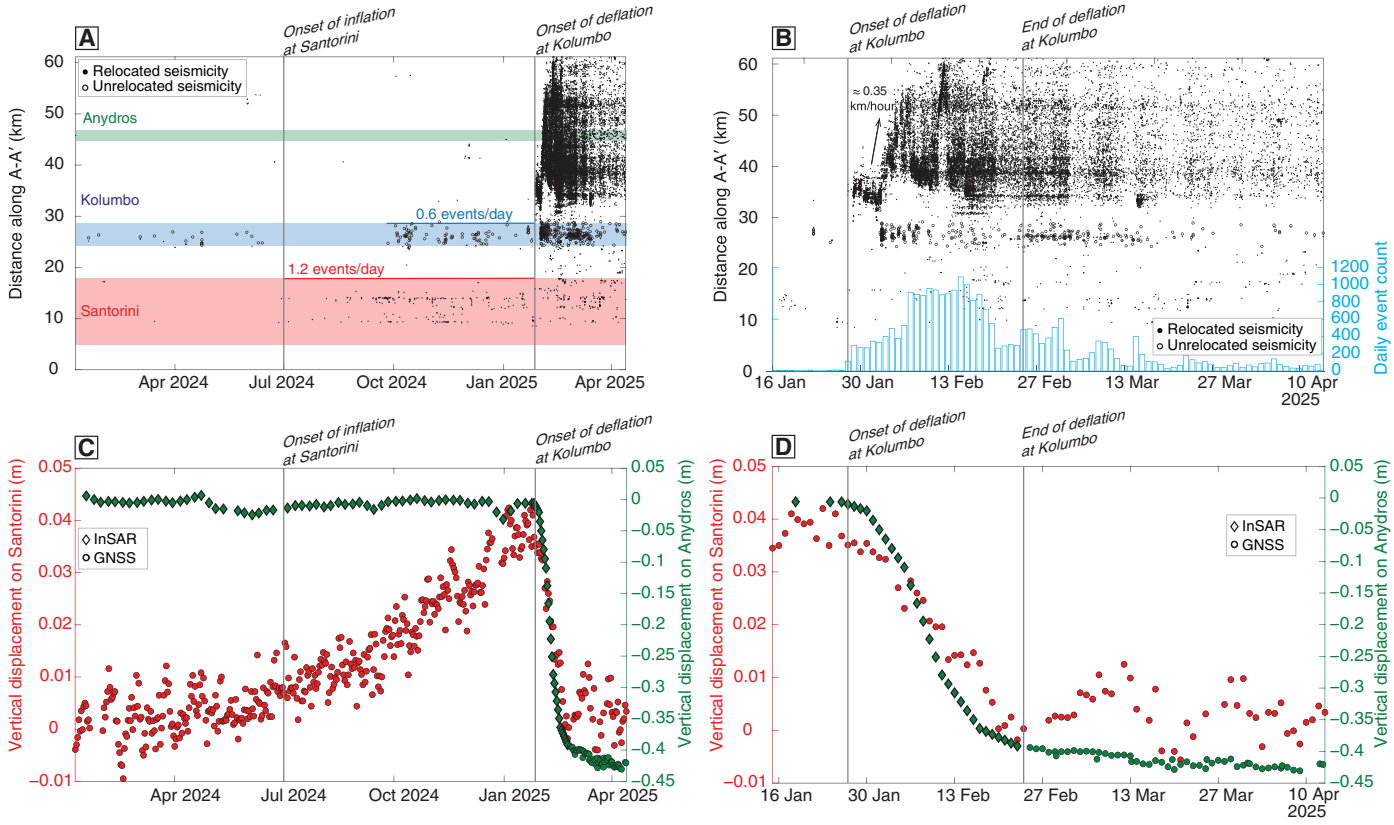


Fig. 2. Evolution of seismicity and deformation spanning the volcanotectonic sequence. (**A**) Location of seismicity along A-A′ as a function of time. (**B**) Magnified view of seismicity in (A) for the period between 15 January and 12 April 2025. The histogram provides daily earthquake counts. (**C**) Vertical displacement time series at the GNSS stations on Santorini (SANT) and Anydros (ANYD). The displacement time series at Anydros is reconstructed by splicing the InSAR time series (colocated with the GNSS station) before 27 January, rescaled InSAR time series (fig. S9) upsampled to daily solutions between 27 January and 24 February, and daily GNSS solutions after 24 February. (**D**) Magnified view of displacement time series in (C) for the period between 15 January and 12 April 2025.

northwest of Nea Kameni (fig. S4 and table S1). Hereafter, we refer to this reservoir as the Santorini reservoir. The reservoir centroid is 5.1 to 5.8 km below the surface, but the aspect ratio of the reservoir is poorly constrained in the inversion (fig. S4C). Our inverted horizontal location for the reservoir centroid (table S1) is similar to that obtained by (*33*, *36*), but our inverted centroid depth and volume increase are larger than those found by (*33*, *36*). This discrepancy likely reflects differences in the procedure of fitting InSAR line-of-sight (LOS) displacements - due to the difficulty of phase unwrapping across the islands, we model LOS displacement as relative displacements (i.e., both observed and predicted displacements are set to zero at an arbitrarily chosen pixel of the displacement map) instead of absolute displacements (see the "Deformation modeling and inversion" section) and differences in parameterization of reservoir geometry (in our inversion, reservoir aspect ratios can vary; see table S1).

The seismic unrest and inflation at Santorini volcano is the first such episode since the January 2011 to March 2012 episode (*37*). If the elastic volume increase in the Santorini reservoir is attributed solely to magma recharge, then the implied magma supply rate is constrained to $\sim 1.9 \times 10^{-2}$ to 2.3×10^{-2} km^3/year, at least an order of magnitude higher than the 1×10^{-3} to 1.6×10^{-3} km^3/year observed during the 2011–2012 episode (*37*) and the average supply rate of 10^{-3} km^3/year in the last 18,000 years (*38*). Alternatively, volume increase could be attributed to second boiling (*39*) (i.e., volatile exsolution from magma due to crystallization). Episodes of transient inflationary deformation are common in caldera systems and do not always lead to eruptions (*40*).

Dike intrusion originating from the Kolumbo volcano

The seismic swarm beginning on 27 January 2025 comprises the majority of the events in our catalog (Fig. 2B). The highest concentration of seismicity extended from ~6 km ENE of the Kolumbo volcano to ~14 km northeast of Anydros and was composed of multiple discontinuous SW-NE trending lineaments of seismicity (Fig. 1, A and B). Most swarm events locate between 8- and 12-km depth (Fig. 1C), although constraints on absolute hypocentral depth are limited in this region because of the poor azimuthal coverage. We also detect shallower seismicity beneath Santorini at depths of ~4 km and beneath Kolumbo volcano at depths of ~7 km (Fig. 1, B and C). Our catalog does not contain the deep seismicity (10 to 15 km) below Kolumbo reported in (*33*). We elaborate on differences between our catalog and existing catalogs in the Supplementary Text.

The onset of the swarm coincided with the onset of deflationary deformation at Santorini (Fig. 2C). During the first 4 days of the swarm, seismicity remained in roughly the same area before migrating to the northeast at ~0.35 km/hour (Fig. 2B). By 2 February, the swarm moved past Anydros before ultimately reaching its maximum spatial extent on 11 February (Fig. 2B).

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Over Santorini, the deflationary deformation (between 27 January 2025 and 24 February 2025) was centered on the northeastern part of Thera (Fig. 1, A and B). Inversion of InSAR LOS displacement and GNSS displacement indicates that the deformation can be explained by a volume decrease of ~0.18 km^3 in a prolate spheroidal reservoir 8.6 km below the Kolumbo volcano (Fig. 3 and table S2). Hereafter, we refer to this reservoir as the Kolumbo reservoir. Volume decrease at the Kolumbo reservoir explains the deformation at Santorini very well (Fig. 3), indicating that concurrent volume decrease at the Santorini reservoir is unlikely. Our location of the centroid of the Kolumbo reservoir is comparable to that found by (*33*), but the volume decrease is more than twice as large as that inferred by (*33*). There are at least three factors potentially contributing to this discrepancy. First, as with the inversion of inflationary deformation at Santorini, our procedure of fitting relative LOS displacements on Santorini and inverting for reservoir aspect ratios are different from that of (*33*). Second, our inversion jointly invert deformation measurements on five islands in the vicinity of the seismic swarm and with magma reservoir, dike, and fault sources, differing from the procedure and data used in (*33*). Last, Isken *et al.* (*33*) used vertical seafloor displacements measured by an ocean bottom sensor, the data of which were not included in our analyses.

The migration of the seismic swarm between 30 January and 3 February toward the northeast (Fig. 2, A and B), the subvertical orientation of the seismicity plane SW of Anydros (Fig. 1D), and the tilt of Ios and Anafi away from the swarm (Fig. 1A) are characteristic

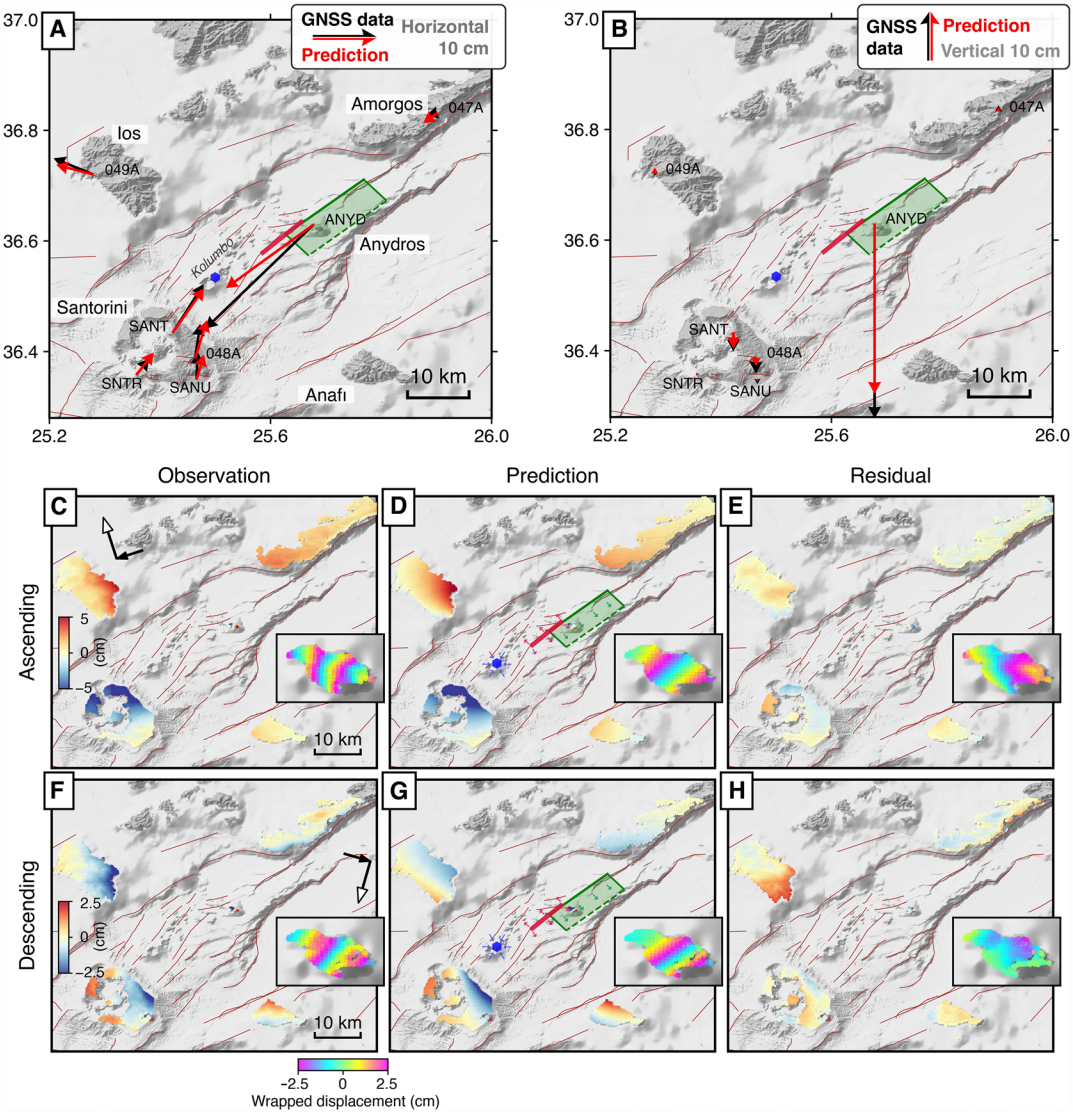


Fig. 3. Observed and modeled deformation during the 2025 deflation-intrusion episode (27 January to 24 February 2025). (**A** and **B**) Comparison of GNSS horizontal and vertical displacements (*35*) with best-fit model predictions. Black and red arrows represent observed and predicted vectors, respectively. (**C** to **E**) Observed, predicted, and residual LOS displacement for the ascending track; (**F** to **H**) corresponding fields for the descending track (note the independent color scales). Orthogonal arrows in (C) and (F) denote the radar flight (open arrow) and positive ground-to-satellite vector (solid arrow). The blue hexagon marks the surface projection of centroid of the Kolumbo reservoir; the red line marks the surface projection of the top edge of the dike; the green rectangle marks the surface projection of the fault (solid and dashed lines indicate the top and bottom edges of the fault, respectively). Insets: Magnified view of displacements on Anydros (wrapped at ±2.5 cm).

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of the lateral propagation of a magma-filled dike. Our inversion indicates that a dike 10.3 km in height, 8.7 km in width, 6.8 km in depth (to the top of the dike), and 5.1 m in average opening can best explain the observed surface deformation between 27 January and 24 February 2025 (Fig. 3). The cumulative dike volume increase amounted to 0.46 km^3. A grid search based on perturbations of parameter values from the best-fit model indicates that the dike width is unlikely to be shorter than 5 km or longer than 12 km (fig. S6). The dike dimensions, orientation, depth, and opening from our inversion (table S2) are similar to those found by (*33*), but our dike is located ~5 km further north compared with that found by (*33*). This discrepancy in dike location is not due to differences in geodetic inversions, but differences in the locations of seismic events (see the Supplementary Text), which are used to anchor the horizontal location of the dike.

The seismicity in our catalog forms a linear trajectory stretching from the Kolumbo volcano to Anydros (Fig. 1, A and B). This geometry, together with the simultaneous onsets of the volume decrease at the Kolumbo reservoir and the dike intrusion (Fig. 2, C and D), suggests that the Kolumbo reservoir was the main source of the dike intrusion along the CSK complex. Intriguingly, the volume decrease at the Kolumbo reservoir between 27 January and 24 February 2025 is more than one order of magnitude greater than the volume increase in the Santorini reservoir between 1 July 2024 and 26 January 2025. The large discrepancy in volume changes between the two reservoirs and the lack of concurrent deflation at the Santorini reservoir during the dike intrusion suggest that the Kolumbo and Santorini reservoirs were hydraulically independent during the 2025 sequence. This raises interesting questions regarding the causal relationship between the pressurization of the Santorini reservoir and the dike intrusion originating from the Kolumbo reservoir, which we address in the "Elastic stress relay enables cascading events" section.

The volume decrease of the Kolumbo reservoir is 39% of the volume increase at the dike. This discrepancy could be explained in terms of the difference in compressibility between the magma reservoir and the dike. Because magma mass, not volume, is conserved and a dike is much more compressible than an ellipsoidal magma reservoir, the volume increase of a dike is typically larger than the volume decrease at the source reservoir (*41*). Assuming a constant magma compressibility β_m, the ratio between the volume increase at the dike and the volume decrease at the reservoir is given by $r_v = 1 + 4\mu\beta_m / 3$ (*41*), where μ is the crustal shear modulus. For $\mu = 20$ GPa, magma compressibility of 5.77×10^{-11} Pa^{-1} explains the observed ratio. This magma compressibility is typical of magmas poor in exsolved volatiles.

Aseismic slip episode on the Amorgos fault

Following the intrusion, seismicity continued throughout the more than 50-km-long swarm. A maximum daily rate of >1000 events/day was reached on 14 February, followed by a slow nonmonotonic decay in seismicity rates until the end of our catalog in mid-April (Fig. 2B).

During the swarm, we observe a remarkable magnitude of deformation over Anydros (Fig. 1, A and B). Starting on approximately January 30 (4 days after deflation began at Kolumbo), a pronounced tilt developed on the 2-kilometer-wide Anydros Island (Fig. 2D), coinciding with the time at which the swarm migrated past the island (Fig. 2B). The western portion of Anydros subsided relative to the eastern portion. Thus, the island was tilting toward the seismic swarm to the northwest. By 11 February, the tilt had produced more than 8.5 cm of LOS displacement across the island (fig. S7), or equivalently, ~8.2 cm of vertical displacement across the island (fig. S8). By 24 February, the vertical GNSS component at the station ANYD (deployed on 12 February) recorded 29 cm of absolute subsidence on the island (*35*). Because the GNSS times series does not cover the entire period of deformation on Anydros, 29 cm is the lower bound in subsidence. Following this episode of rapid deformation, the rate of deformation decayed substantially (Fig. 2D). We estimate that, by 24 February 2025, the differential vertical displacement across Anydros is ~18 cm and the absolute subsidence at the site of GNSS station ANYD is ~39 cm. The absolute subsidence is derived by assuming a four-and-half-to-one subsidence-tilt relation by comparing InSAR-derived differential displacement with the GNSS displacement at ANYD for the period between 12 February and 24 February 2025 (see the "Estimating time-dependent aseismic and seismic moment" section; fig. S9). Given the noise level in the InSAR data and the lack of preunrest trend calibration for the GNSS record, we cannot determine whether the deformation on Anydros has ceased or is still ongoing at a low level as of September 2025 (figs. S7 and S8).

We find that dike intrusion in the absence of concurrent aseismic faulting cannot simultaneously explain the large magnitude of subsidence and the gentle tilt of Anydros toward the northwest (fig. S5). For example, a dike with its bottom edge fixed at 10 km below the surface and its top edge shallower than 5 km fits neither the subsidence nor the tilt at Anydros for any amount of dike opening. For a dike with its bottom edge fixed at 10 km and its top edge deeper than 5 km, only unrealistic amounts of opening (>100 m) can explain the 39 cm of subsidence at the GNSS station ANYD. Even in these unrealistic scenarios, the model overpredicts the tilt over the island by at least a factor of three (fig. S5). A dike intrusion where the northeastern end is significantly shallower (fig. S5), such as considered by (*33*), does not qualitatively change this assessment. Thus, an additional deformation source is needed to explain the subsidence and tilt at Anydros.

The SE-dipping structures outlined by seismicity northeast of Anydros (Fig. 1E) strongly suggest that aseismic faulting is one of the underlying mechanisms for the deformation at Anydros. We find that the Anydros deformation (between 27 January and 24 February 2025) can be explained by 0.56 meters of dip slip and 0.25 m of dextral slip on a fault 16.8 km in along-strike length, 9.6 km in along-dip length, and 2.1 km in depth to its upper edge (Fig. 3 and table S3). The fault dips to the southeast at 54°, consistent with seismicity locations northeast of Anydros (Fig. 1E). The dip and the dimensions of this fault are strongly constrained by the GNSS displacements (fig. S6). Additionally, the slip magnitude ($D = 0.62$ m) and the size of the fault ($A = 161.3$ km^2) from inversion of geodetic data are consistent with a self-similar earthquake ($D \sim \Delta\tau A^{1/2} / \mu$; for a typical stress drop $\Delta\tau$ of 1 MPa and shear modulus μ of 20 GPa), although we do not enforce self-similarity in the inversion. Using the vertical displacement at Anydros (fig. S9) as a proxy for the time dependence of the aseismic slip transient, we found that the aseismic slip likely started on 30 January and ended on 24 February (Fig. 2D), coinciding with the end of volume increase at the dike.

The fault plane from our inversion is distinct from that proposed by (*35*). The best-fit fault plane found in (*35*) is located south of Anydros. The difference between our fault plane models is likely

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attributable to our accounting of the dike intrusion and volume decrease of the Kolumbo reservoir for the inversion. Our best-fit fault model closely aligns with a SE-dipping branch of the Amorgos fault northwest of Anydros (Fig. 1, A and B), previously imaged in seismic reflection surveys (*42*, *43*). Dip slip and dextral slip on a SW-NE branch of the Amorgos fault would be consistent with the right-lateral transtensional tectonic environment of the CSK complex (*44*).

DISCUSSION

Elastic stress relay enables cascading events

The seismic crisis of 2025 spanned more than 50 km along the CSK complex, a spatial scale much larger than that of any individual magma reservoir, dike, or fault in the complex (Fig. 4A). During the period, the largest seismic event was M_w 5.2, unlikely to affect major elastodynamic stress perturbations along the CSK complex. Elastostatic stress perturbations, however, scale with the reciprocal of the distance to the source cubed, so the stress perturbation due to each deformation source is negligible at a distance of a few source dimensions away. For example, the pressurization of the Santorini reservoir produces a tensile stress perturbation of order 10^{-3} MPa on the Amorgos fault (Fig. 4A), roughly an order of magnitude smaller than typical ocean tide loading.

We postulate that the sequential triggering of each intrusion/faulting event by its preceding, neighboring event is likely crucial for long-distance interactions in the rift and is enabled when the crust is sufficiently close to shear and/or tensile failure. We term this mode of long-distance volcanotectonic interaction elastic stress relay to distinguish this behavior from pairwise, short-distance stress interactions [e.g., (*5*, *9*, *11*, *13*, *45*, *46*)].

We infer that the pressurization of the Santorini reservoir in mid-2024 facilitated the dike intrusion originating from the Kolumbo reservoir, thus initiating the 2025 volcanotectonic sequence. Two lines of evidence support this scenario. First, stress modeling (Fig. 4A; see the "Perturbation tractions and Coulomb stress" section) indicates that pressurization of the Santorini reservoir applied a perturbation tensile stress of ~0.1 MPa at the Kolumbo reservoir, where the dike intrusion initiated. Such perturbation stress can be further amplified by a factor contingent upon the geometry of the reservoir (*47*, *48*), given that magma reservoirs are stress concentrators in the crust (consider the well-known problem of a cylindrical hole in a two-dimensional (2D) elastic plate, for which the uniaxial tensile stress in the far field is amplified by a factor of three at the edge of the hole). A slightly elongated Kolumbo reservoir along the rift, as suggested by our inversion (table S2), may further enhance the likelihood of dike initiation due to pressurization of the Santorini reservoir. For reservoir wall rocks under tensile loading in a rift environment, it is conceivable that tensile stress perturbations of this magnitude are sufficient to initiate dike intrusion. Second, the GNSS time series at station SANT on Santorini (Fig. 5A) indicates that the dike intrusion was unlikely to be caused by a sudden surge in magma supply to the Kolumbo reservoir. During the 2025 unrest, ~5 cm of eastward displacement was recorded at SANT due to the volume reduction of 0.18 km^3 at the Kolumbo reservoir. Assuming linear elasticity, we expect ~5 cm of westward displacement in the precrisis period if all the magma volume lost at the Kolumbo reservoir in 2025 was accumulated in the 10-year period preceding the crisis. However, given that there is no discernible displacement at SANT in the precrisis period (Fig. 5A) and that typical uncertainties of horizontal GNSS displacement are 1 to 3 mm, we argue that no more than $^1/_{50}$ to $^1/_{16}$ of the magma volume lost in Kolumbo reservoir in 2025 was added to the reservoir in the preceding 10 years. Beyond elastic stress interactions, the Santorini reservoir and the Kolumbo reservoir were also unlikely to be hydraulically connected during the 2025 unrest. As indicated by the continued inflation of the Santorini reservoir post-intrusion (Fig. 5, B and C), the magma supply to the Santorini reservoir was not substantially affected by the dike intrusion.

Elastostatic stress interactions between reservoirs are rarely considered for modeling volcanic hazards, yet there is overwhelming geophysical evidence that magmatic reservoirs often exist in clusters [e.g., Kīlauea (*15*), Etna (*16*), Axial Seamount (*17*), Krafla (*18*), and Hunga Tonga (*14*)] or transcrustal systems (*49*). The potential for interactions in these systems will be informed, in each setting, by the criticality of the stress state and the geometry of the reservoirs. Future studies should evaluate how volcanic risks compound in such scenarios.

The oblique dip-slip faulting on the SW-NE trending section of the Amorgos fault was likely triggered by the dike intrusion. We estimate that, by 24 February 2025, the dike would have produced ~1 MPa of Coulomb stress change on the Amorgos fault in the dip-slip direction (Fig. 4, A and B; see the "Perturbation tractions and Coulomb stress" section). However, at the time of slip initiation on the Amorgos fault (around 30 January), the perturbation stress on the fault was likely smaller. The initiation of shear faulting with <1 MPa of Coulomb stress change would be consistent with critically stressed fault systems in the transtensional CSK complex. In theory, because slip on the Amorgos fault occurred concurrently as the dike intrusion, the shear slip should also induce normal traction perturbations on the dike. A scenario in which the dip-slip faulting prevented the dike from propagating further northeast should be tested via stress modeling, but is beyond the scope of this project.

The numerous seismic events up to M_w 5.2 during the 2025 seismic crisis, therefore, are driven by both dike intrusion and aseismic faulting. In reality, aseismic faulting needs not be restricted to a single fault plane, such as in our geodetic inversion (Fig. 3) but could be distributed across a system of subparallel faults northeast of Anydros (Fig. 1E). One example in support of distributed aseismic faulting is the occurrence of a burst of seismicity migrating northeastward on 10 February 2025 (Fig. 6A). This burst of seismicity comprises two subparallel lineaments migrating laterally at ~12 to 14 km in depth (Fig. 6B), below the geodetically determined slip surface (the bottom edge of which is at a depth of 9.9 km). Temporally, the seismicity burst comprises three subbursts (Fig. 6C), each of which originated at the tip of the preceding burst. This pattern is reminiscent of sequential triggering of aseismic faulting on a series of normal faults during the 2000 Miyakejima dike intrusion (*12*). Therefore, we infer that the seismicity burst was driven by northeastward sequential ruptures of subparallel faults in a similar orientation as the main slip surface of the Amorgos fault.

Aseismic slip transients and implications for seismic hazards

Our results (Fig. 3; see the "Aseismic slip episode on the Amorgos fault" section) suggest that aseismic fault slip is a previously unrecognized driver of seismic hazards in the CSK complex. During the 2025 crisis, the cumulative aseismic moment due to

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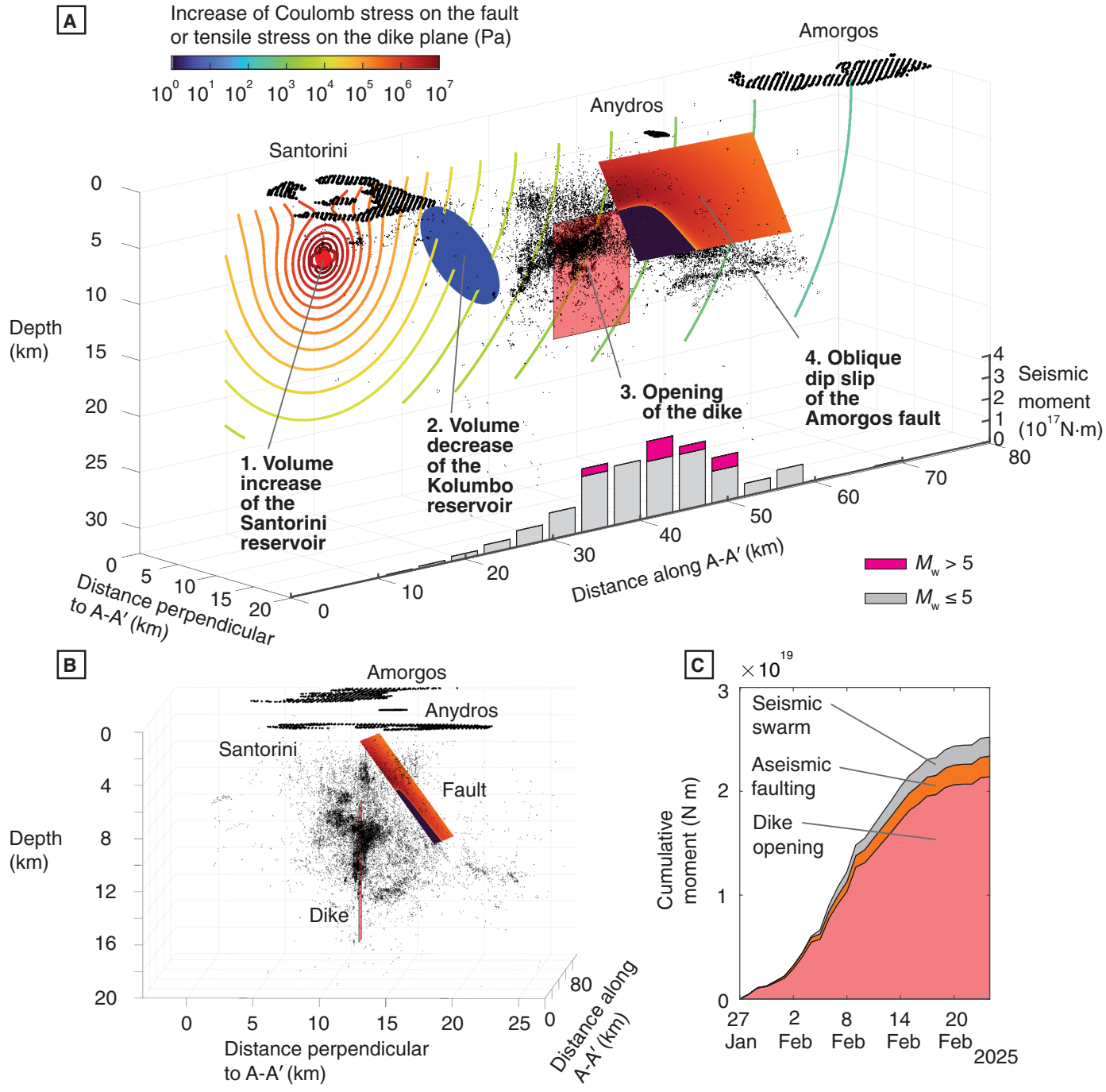


Fig. 4. Cascading dike intrusion and aseismic slip transient enabled by elastic stress relay. (**A**) Coulomb stress increase on the Amorgos fault (only nonnegative Coulomb stress changes are shown) due to dike opening and tensile stress increase on the dike plane due to volume increase at the Santorini reservoir. Seismic events are marked by black dots and their moment distribution along A-A′ is plotted as a histogram. All deformation sources and receivers for stress calculations and for plotting are based on the best-fit model of our inversions (tables S1 to S3). The dimensions of the Santorini reservoir and the Kolumbo reservoir are weakly constrained by inversions. (**B**) A view of the dike and the fault looking northeast. (**C**) Cumulative seismic and aseismic moment as a function of time.

dike intrusion amounts to 2.14×10^{19} N·m or 84.9% of total moment release (Fig. 4C; see the "Estimating time-dependent aseismic and seismic moment" section). Aseismic faulting and the seismic swarm account for 7.8 and 7.3% of total moment, respectively (Fig. 4C; see the "Estimating time-dependent aseismic and seismic moment" section).

The aseismic slip transient (27 January to 24 February 2025) observed in the CSK complex may be explained by the same frictional regimes that allow slow slip events to occur: the strengthening of friction through either changes in rate and state parameters (*50*) or dilation of fault gouge (*51*) with increasing slip rate. Slow slip events, however, nucleate and propagate under steady tectonic

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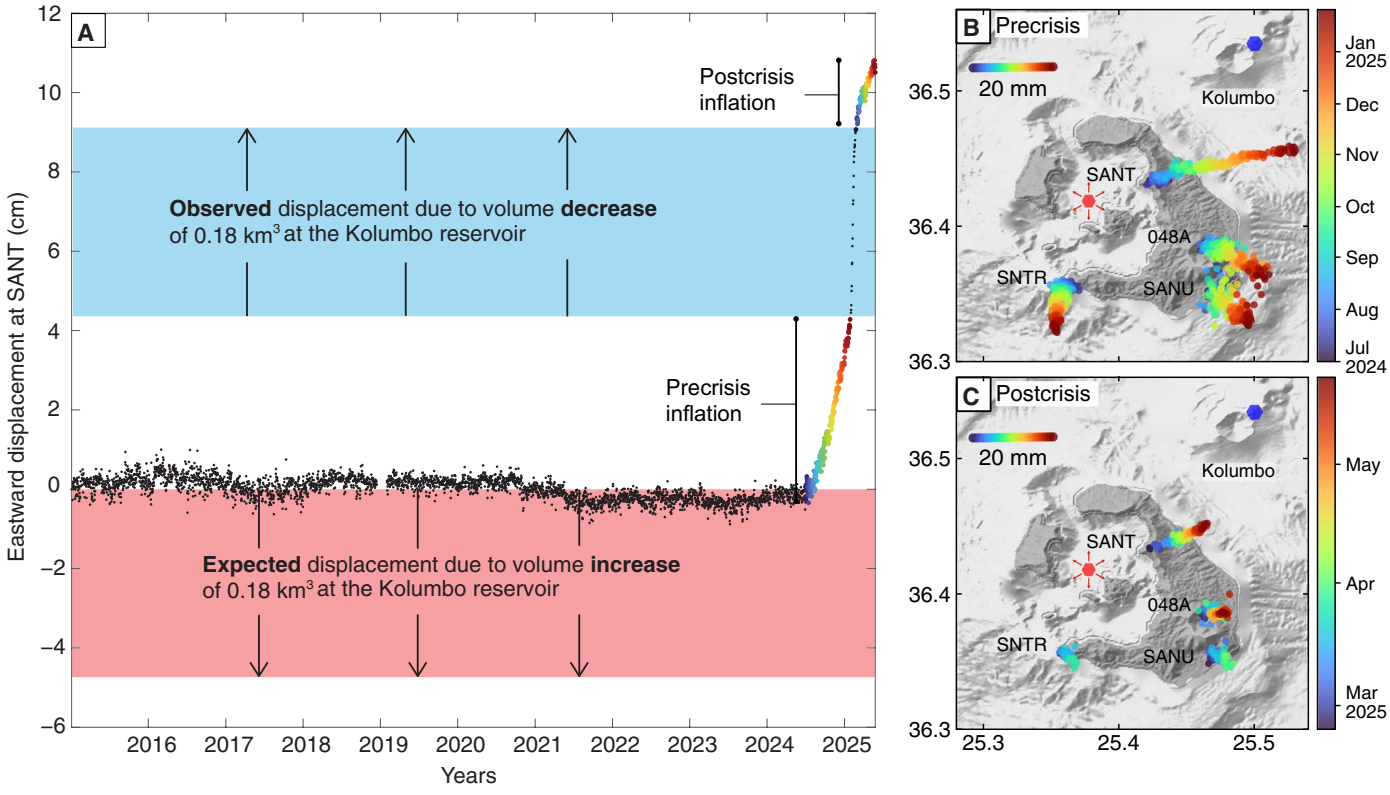


Fig. 5. Absence of inflation at Kolumbo before the 2025 seismic crisis and the continuation of inflation at Santorini after the 2025 seismic crisis. (**A**) GNSS eastward displacement at station SANT on Santorini, covering an ~10 year period leading up to the 2025 crisis. Inflationary deformation in the precrisis (from 1 July 2024 to 26 January 2025) and postcrisis periods (from 23 February 2025 to 23 May 2025) are color coded with time to match (B) and (C). Blue shading denotes the observed eastward displacement due to volume decrease at the Kolumbo reservoir. Red shading indicates the expected westward displacement for an equivalent volume increase at the Kolumbo reservoir, assuming linear elasticity. (**B**) GNSS displacements during the 210-day precrisis period. (**C**) Corresponding displacements during the 90-day postcrisis period. The red and blue symbols denote the centroids of the Santorini and the Kolumbo reservoirs, respectively.

loading, whereas the aseismic slip transient on the Amorgos fault was driven by external stress perturbations from the dike intrusion. Differences in the driving mechanisms may account for the longer duration and larger slip magnitude of aseismic slip transients observed on the Amorgos fault, compared with typical slow slip events in volcanic regions and subduction zones. For example, slow slip events on the décollement fault of Kīlauea have durations between 1.5 and 2.2 days and a slip magnitude of centimeters (*52*). Slow slip events in subduction zones have durations more consistent with our observations (*53*), but the typical slip magnitudes in subduction zones are on the order of centimeters, much smaller than our inferred aseismic transient. The closest example to our observation may be the dike intrusion-induced aseismic slip at Miyakejima in 2000, which lasted tens of hours and reached a slip magnitude of ~0.4 m (*12*).

Our observed aseismic slip transient raises important questions regarding the seismic hazards in the CSK complex. The geodetically determined principal slip surface (Fig. 3) coincides with the southwestern end of the Amorgos fault, which hosted the 1956 Amorgos earthquake (M_w 7.8), the largest earthquake in Greece in 20th century. However, our geodetic inversions lack the resolution to determine an upper bound on the SW-NE extent of faulting (fig. S6); therefore, the spatial relationship between the aseismic slip transient identified in this study and the 1956 earthquake remains unclear. Additional seismic and geodetic monitoring of the Amorgos fault in the future may be warranted.

Segmented dike growth via magma pulses

During the swarm, several concentrated bursts of seismicity migrated at kilometer-per-hour rates within the geodetically determined dike (Fig. 7A). In terms of depths, the seismicity bursts situate in the upper half of the geodetically inferred dike (Fig. 7B). The seismicity bursts also coincide with a proposed midcrustal magma reservoir associated with a large low-V_p anomaly 8 to 12 km in depth (*27*), although the relationship between the proposed reservoir and the seismicity bursts are unclear.

In contrast to the horizontally propagating seismicity driven by aseismic fault slip (Fig. 6), each burst of seismicity along the dike migrated predominantly upward from its point of initiation and was confined within the dike plane (Fig. 7B). Each seismicity burst lasted between 1.6 and 3.8 hours and traversed a vertical distance of 2 to 6 km.

The coherent pattern of upward-migrating seismicity suggests that opening along the dike plane drives the seismicity: Shear ruptures are unlikely to propagate unidirectionally upward, absent of idiosyncratic stress or frictional heterogeneities. A parsimonious explanation is that each burst of seismicity represents a magma pulse, which we define as a small dike extending from the top of a

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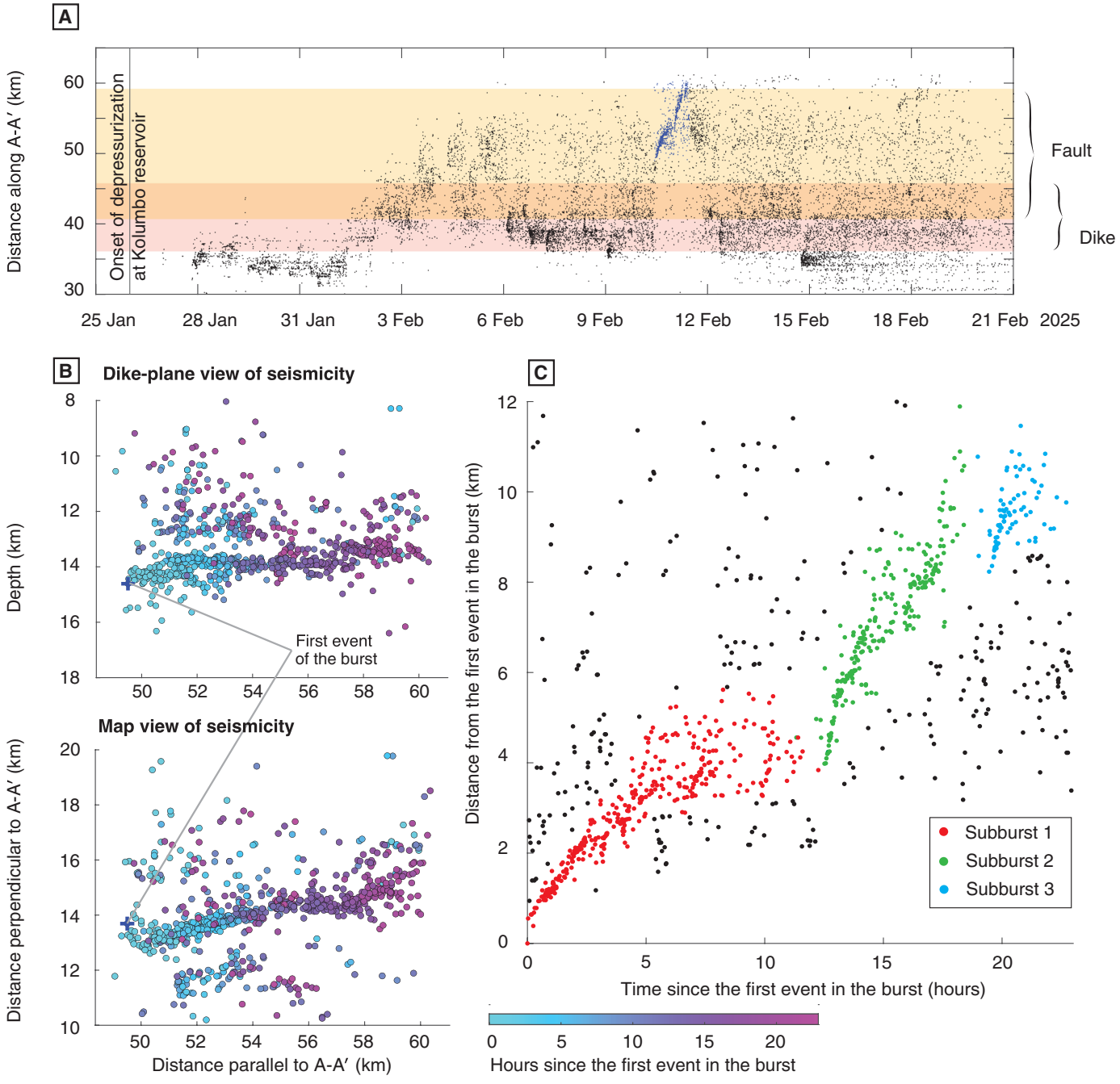


Fig. 6. Seismicity associated with distributed aseismic slip. (**A**) A seismicity burst (highlighted in blue) migrating at kilometer-per-hour rates along the strike of the geodetically determined fault (Fig. 4). (**B**) Magnified view of the seismicity burst, which located ~2 to 4 km below the geodetically determined fault. The seismicity burst predominantly propagated toward the northeast. (**C**) Distance-time plot of the seismicity burst, showing three distinct subbursts. Each subburst initiated at the edge of the previous subburst, characteristic of aseismic faulting sequences across multiple faults (*12*).

major dike to promote laterally segmented opening. This conceptual model is consistent with well-documented field exposures of segmented dikes (*54*), which commonly occur in the presence of preexisting faults and mixed deviatoric-tensile stress fields. Both of these conditions are abundantly present in the CSK complex.

The upward migration of each magma pulse, together with effectively constant along-dike dimension or breadth (Fig. 7B), strongly indicates that propagation was driven by buoyancy; a magma pulse driven by magma overpressure stops growing radially when buoyancy dominates over the pressure gradient due to magma reservoir overpressure (*55*). The propagation of a constant-volume, buoyancy-driven magma pulse (much like a dike) is controlled by the balance between the buoyancy force due to the density difference between the magma and the host rock and the viscous force resisting the flow

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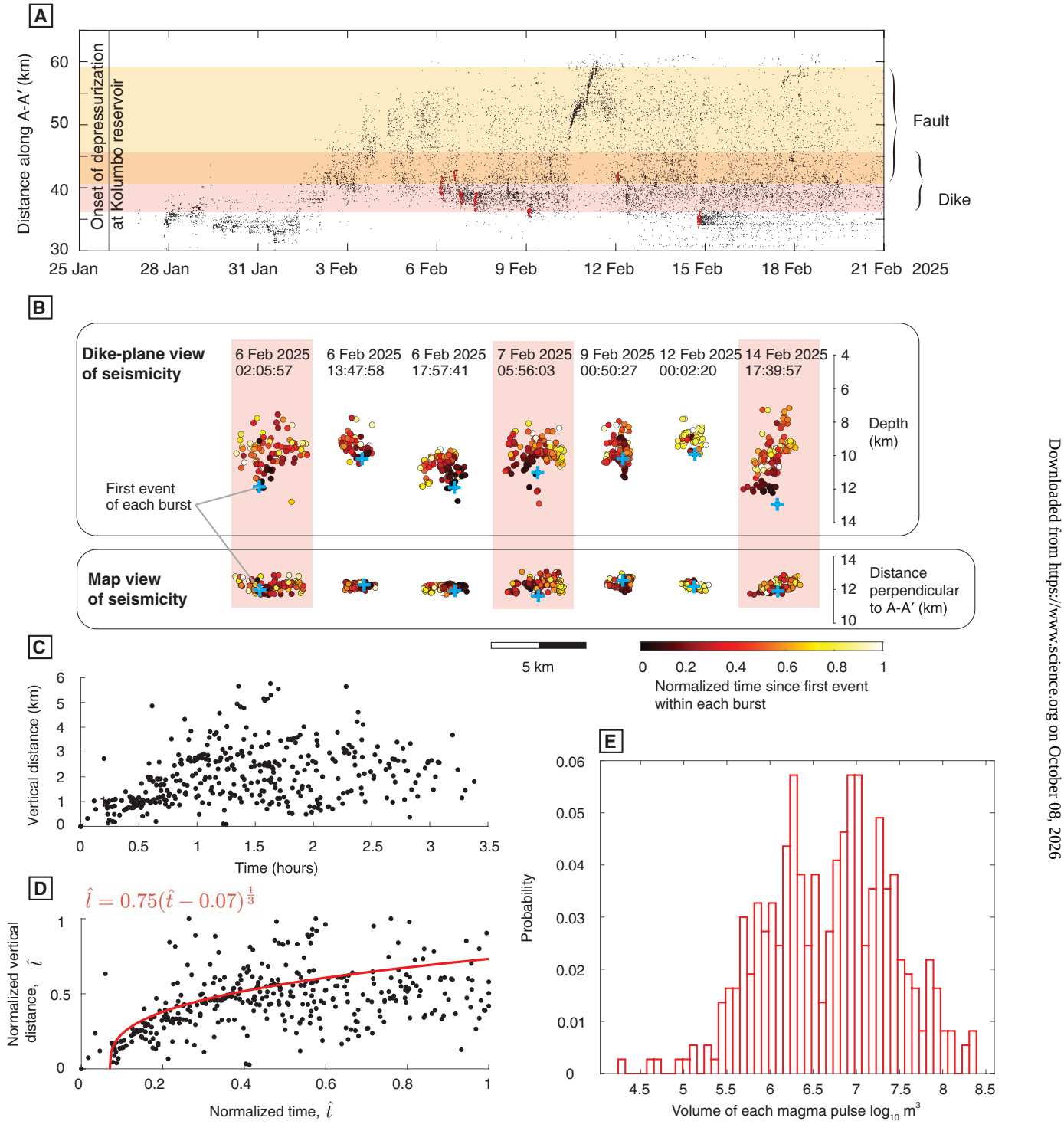


Fig. 7. Seismicity associated with segmented, upward dike growth. (**A**) Seismicity bursts (highlighted in red) migrating at kilometer-per-hour rates within the geodetically determined dike. (**B**) Magnified view of seven representative seismicity bursts, which predominantly propagate upward and are characteristic of buoyancy-driven magma pulses. (**C**) Vertical distance of each seismic event (relative to the first event within each burst) with time, aggregated from three representative bursts highlighted in (B). (**D**) Same as (C), with distance normalized by the maximum vertical distance of each pulse and time normalized by the maximum duration of each pulse. The implied magma pulse height follows a power law obtained from the balance of magma buoyancy force and viscous force. (**E**) Volume of each magma pulse inferred from a Monte Carlo simulation of magma viscosity, magma density difference from host rock, the observed range of magma pulse breadths, and the height of magma pulses as a function of time.

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of magma. This balance leads to a power-law scaling between the height of the pulse, l, and time, t (*56*)

$$l = \left(\frac{\Delta\rho g V^2}{4\eta B^2} t\right)^{1/3} \quad (1)$$

where $\Delta\rho$ is the density of the crust less that of the magma, η is the dynamic viscosity of the magma, and V is the constant volume of the magma pulse. g is the gravitational constant, and B is the constant breadth of the magma pulse. The propagation rate of the magma pulse decreases as the magma pulse grows because the viscous force increases with the magma pulse area (or roughly the square of magma pulse height), whereas the buoyancy force increases linearly with magma pulse height.

We verify that observed magma pulses approximately follow the power-law scaling of Eq. 1 (Fig. 7, C and D). Power-law relationships lack a characteristic length scale separating regimes, which suggests self-similarity of the pulse height with respect to time. To see this, we nondimensionalize variables l and t in Eq. 1 with length scale L and the corresponding time scale T, obtaining

$$\hat{l} = \hat{t}^{\frac{1}{3}} \quad (2)$$

where $\hat{l} = l/L$, $\hat{t} = t/T$, and L and T are theoretically related as $L = \left(\frac{\Delta\rho g V^2}{4\eta B^2}\right)^{\frac{1}{3}} T^{\frac{1}{3}}$. Equation 2 suggests that rescaling magma pulse height (defined as the vertical distance from the initiation point of each burst to the seismic event farthest away at given time) by its final height, L, and rescaling propagation time (referenced to the initial event of each burst) by the final duration of the pulse, T, should cause the depth histories of each pulse to collapse onto a single curve. We show that (Fig. 7, C and D), after applying a small correction for bias in estimated initiation time of each burst (hence, the time shift of 0.07 applied to $\hat{t}$) and bias in the estimated height of the magma pulse (hence, the scaling factor of 0.75), the observed seismicity bursts follow a power law given by Eq. 2.

We estimate the volume of each magma pulse by Monte Carlo sampling of the plausible range of parameters in Eq. 1. We choose seismicity bursts with clear migration patterns (those highlighted in Fig. 7B) for this analysis. We treat each seismic event as an independent measurement of magma pulse height, l, at time, t, and estimate $\left(\frac{\Delta\rho g V^2}{4\eta B^2}\right)^{\frac{1}{3}}$ from $l/t^{1/3}$. Values of B are sampled between 2 and 4 km, reflecting the observed breadths of seismicity bursts. Values of $\Delta\rho$ and η are sampled between 0 and 400 kg m^{-3} and between 10^1 and 10^5 Pa · s, respectively, ranges appropriate for basaltic magma (magma with higher viscosity precludes magma pulses propagating at kilometer-per-hour rates, such as observed here). All sampled parameters are assumed to have uniform probability distributions within the given bounds.

We determine that each magma pulse has a volume of 9.7×10^5 to 3.2×10^7 m^3 at a 68% confidence interval (Fig. 7D). Therefore, each magma pulse is accompanied by decimeter- to meter-scale opening (obtained by taking the volume of each magma pulse and dividing by its area, as indicated by seismicity), in agreement with the idea that each magma pulse represents an increment of opening during segmented dike growth.

In conclusion, we identify a cascading sequence of unrests from the seismicity and deformation data recorded during the 2025 volcanotectonic sequence in the CSK complex. We infer that, the pressurization of a magma reservoir beneath Santorini over a 7-month period imparted >0.1 MPa of tensile stress increase to a reservoir beneath Kolumbo. As a result, the wall rocks of the reservoir under Kolumbo failed in late January 2025 and produced a month-long dike intrusion. This dike then triggered a M_w 6.2, aseismic, oblique dip-slip transient on a previously mapped branch of the Amorgos fault. Numerous seismic events up to M_w 5.2 were triggered by the combined stressing of dike intrusion and oblique dip-slip faulting. This relay of elastostatic stress enabled the propagation of volcanotectonic unrest over a more than 50-km section of the CSK complex, much longer than the dimension of each individual magma reservoir and fault comprising the complex. Within the upper portion of the geodetically determined dike, we identify bursts of rapid, upwardly propagating seismicity, which we interpret as buoyancy-driven magma pulses promoting segmented dike growth. Our work demonstrates that developing an accurate understanding of eruptive and seismic hazards in rift zone systems requires spatially extensive, multidisciplinary monitoring networks that can observe processes far beyond the footprint of a single volcano or fault.

MATERIALS AND METHODS

Seismicity catalog construction

To construct a seismicity catalog for the swarm, we download continuous waveform data spanning 1 January 2024 to 11 April 2025 from the National Observatory of Athens (NOA) webservice for nearby stations in the GE, HA, HI, HL, and HT networks (fig. S1) (*32*). We perform phase picking on the time-series data using the automated phase-picking platform SeisBench (*57*), using the standard PhaseNet model. PhaseNet requires a user-input threshold value between 0 and 1 to differentiate true and false phase detections. Because the azimuthal coverage of our network is poor, with nearly all nearby stations being located on Santorini (fig. S1), we elect to use a low detection threshold of 0.1 for both P and S picks to retrieve as many true detections as possible. We recover 3,889,195 P and S picks from the data.

Next, we associate these picks using the phase association software PyOcto (*58*). PyOcto performs phase association by backpropagating detected phases through a user-defined 1D velocity model. We construct a 1D velocity model for the region by taking average values at each depth slice of a recently published local 3D crustal model (*27*). We require a minimum of eight picks and at least two stations with both a P and S pick to define an event and recover 81,589 events.

Last, we locate the events using a modified version of the earthquake location software HypoSVI (*59*). Typically, HypoSVI locates events by minimizing travel-time residuals using a physics-informed neural network as an eikonal equation solver. Due to the complex 3D velocity structure of the crust, we replace the neural network forward model with interpolable travel-time tables calculated using the velocity model of (*27*).

Because the azimuthal coverage of available seismic stations is poor, accurately constraining absolute hypocenters in the swarm area is challenging. Additionally, because of the low detection threshold we use for PhaseNet, it is also likely that our pick dataset contains outliers or false detections that may bias locations. To combat the effects of spurious travel-time data and limited azimuthal coverage, we use a two-step method to construct our catalog. First, we

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calculate locations using an iterative source-specific station term (SSST) process by (i) determining an initial location for each event, (ii) calculating a static station correction term for each station-phase pair and locating events again with these correction terms applied, and (iii) calculating SSSTs for each event by taking the median of the residuals of the 250 closest events, applying these to travel-times as correction terms, and updating our locations. The median absolute deviation (MAD) of travel-time residuals decreases as expected with each iteration of the source-specific corrections (fig. S2). We continue step (iii) for three iterations, at which point the correction terms and residual MAD become relatively stable.

Next, we use this set of locations to identify outlier picks. We identify and discard outliers by calculating the MAD of the travel-time residuals for each station-phase pair and removing picks with an absolute residual greater than four times the MAD (fig. S2). We also remove events that have fewer than eight picks after this filtering process. We then set the SSSTs to zero and repeat the location process with the outlier picks removed, calculating one set of static correction terms followed by four sets of source-specific terms. At the end of this process, we remove events with retrieved depths <0 or >50 km relative to sea level. The resulting catalog contains 63,931 events and displays markedly improved location clustering and residual distribution (residual MAD, 0.114 s) compared with the starting catalog without outlier removal or travel-time corrections (residual MAD, 0.254 s) (fig. S2).

Last, we relocate our catalog using the cross-correlation earthquake relocation package GrowClust (*60*). To perform waveform relocations, we filter event waveform data from 1 to 20 Hz and cross-correlate each event's *P* and *S* arrivals with those of the nearest 5000 events. We require a minimum cross-correlation threshold of 0.7 and require a minimum of six differential times for an event to be relocated. We successfully relocate 25,227 events (Fig. 1). We consider the relocated events to be the final version of our catalog and use these events in subsequent plots, except where noted. We note that our catalog is slightly temporally biased because of cultural noise during local daytime on Santorini (fig. S3).

InSAR processing

Satellite radar acquisitions are obtained from the European Space Agency Sentinel-1 satellite. We download two tracks of images (A029 and D109) from the Alaska Satellite Facility DAAC archive spanning January 2024 to July 2025. Using the InSAR Scientific Computing Environment [ISCE2; (*61*)] software, we coregister and process radar images into interferometric pairs in a small-baseline subset fashion, with a maximum temporal baseline of 120 days (corresponding to the ten nearest consecutive acquisitions at a nominal 12-day cadence). Eighteen months of interferograms are multilooked to a 30-m ground spacing, filtered, unwrapped, and geocoded (*62*, *63*). The data metrics and processing hyperparameters are listed in table S4.

We spatially crop the datasets to reference and reconstruct LOS displacement time series for five regions: Santorini, Anydros, Ios, Amorgos, and Anafi. For each region, we discard interferograms with an averaged coherence lower than 0.7 from the interferometric network. To mitigate phase unwrapping errors, we use a minimum-spanning-tree algorithm to bridge the phases across the several islands of Santorini (Thera, Nea Kameni, Palea Kameni, and Therassia) for each interferogram (*64*). The amount of integer-misclosure triplets (triplet closure phase > 2π) in the network is reduced from ~80% to less than 10% with phase bridging. Residual phase artifacts are modest and further suppressed after the time-series inversion flooded with high-quality interferograms. We further mask out pixels with temporal coherence lower than 0.7 (near steep caldera cliffs and Mesa Vouno). The rest of the islands (Ios, Anydros, Amorgos, and Anafi) are processed in the same way but the reference points are chosen within its own realm (table S5), rather than bridged to Santorini. We account for phase contributions from troposphere delay (ERA5–European Center for Medium Range Weather Forecasts Re-Analysis model and additional empirical height correction), solid Earth tides, and baseline-related topographic residuals. Given the limited spatial scale of the study area, ionospheric effects are neglected. The apparent LOS velocity associated with Eurasia's secular plate motion relative to ITRF2014 (the International Terrestrial Reference Frame) is also negligible (0.2 mm/year across 10 km).

GNSS and InSAR time-series analysis

In general, time-dependent displacements may be expressed as a superposition of temporal basis functions (*65*). To represent the transition from inflation to deflation at Santorini, we adopt the following time-dependent components for InSAR displacement time series: a background secular trend to account for regional tectonics before July 2024, an inflationary trend active from July 2024 until the dike intrusion, and a subsequent deflation-intrusion trend initiated as a Heaviside step on 27 January 2025. For each pixel, the LOS displacement $d(\mathbf{x}, t)$ is thus parameterized as

$$d(\mathbf{x},t) = v(\mathbf{x})t + v_{\text{infl}}(\mathbf{x})\left(t - T_{\text{infl}}\right) + S(\mathbf{x})H\left(t - T_{\text{defl}}\right) + \sum_i \alpha_i(\mathbf{x})H\left(t - T_{\text{defl}}\right)\ln\left(1 + \frac{t - T_{\text{infl}}}{\tau_i}\right) \quad (3)$$

The background trend is $v(\mathbf{x})$. The inflation in 2024 is fitted with a linear velocity v_{infl} starting at $T_{\text{infl}} = 1$ July 2024. The deflation and dike intrusion began at T_{defl} and is modeled as a Heaviside step $S(\mathbf{x})$ plus logarithmic relaxations $\alpha_i(\mathbf{x})$ with times τ_i. On Santorini and Ios, deflationary deformation started as early as 27 January, followed by other islands (table S6). A two-term relaxation with $\tau_1 = 1$ day and $\tau_2 = 10$ days approximately fits the observed transients across all regions.

Outside the Santorini region, inflationary deformation associated with the Santorini reservoir is absent. Only Santorini and Anydros require step functions to capture the rapid onset of the deflationary deformation. For Ios, Amorgos, and Anafi, the deformation during the deflation period can be explained with logarithmic terms only. To minimize trade-offs among basis functions, the logarithmic components are constrained to maintain consistent polarity on each island. Net deflation is quantified by differencing the predicted displacements between the final epoch and the epoch immediately preceding the intrusion and seismic swarm. These basis functions are not intended to represent distinct physical processes, but rather to serve as mathematical filters that isolate the dominant, smoothed temporal patterns.

GNSS displacement time series is taken directly from (*35*). For intuition, radar LOS displacements are reprojected back into nominal east, north, and up components. With only ascending and descending tracks, the 3D decomposition is underdetermined; one component lies in the null space. We therefore estimate horizontal u_h and vertical u_z by combining the two geometries, constraining the horizontal direction to the GNSS azimuth.

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$$\begin{bmatrix} d_{\mathrm{asc},i} \\ d_{\mathrm{dsc},i} \end{bmatrix} = \begin{bmatrix} \sin\theta_{\mathrm{asc},i}\cos\left(\phi_{\mathrm{asc},i} - \alpha_{\mathrm{GNSS}}\right) & \cos\theta_{\mathrm{asc},i} \\ \sin\theta_{\mathrm{dsc},i}\cos\left(\phi_{\mathrm{dsc},i} - \alpha_{\mathrm{GNSS}}\right) & \cos\theta_{\mathrm{dsc},i} \end{bmatrix} \begin{bmatrix} u_{h,i} \\ u_{z,i} \end{bmatrix} \quad (4)$$

where $d_{\mathrm{asc},i}$ and $d_{\mathrm{dsc},i}$ are the ascending and descending LOS displacements at pixel i, with incidence and azimuth angles $\theta_{_,i}$ and $\phi_{_,i}$. The parameter α_{GNSS} is fixed as the azimuth of the GNSS horizontal motion on each island in the deflationary period [table S4 of (*35*)]. Thus, $u_{h,i}$ and $u_{z,i}$ denote InSAR-derived horizontal displacement along the GNSS azimuth and the vertical component.

Figures S7 and S8 present the GNSS displacement time series, InSAR LOS displacement time series, and the horizontal and vertical components resolved from LOS displacement. GNSS data are referenced to ITRF2020; we subtract a secular velocity, estimated from pre-2024 data, to isolate transient deformation from other long-term tectonic effects [with the exception of station ANYD due to its incomplete data record; (*35*)]. InSAR displacements are represented by the median value within a 1 km^2 window, with the $\pm 1\sigma$ spread indicating spatial variability. These windows are centered on GNSS stations where available; otherwise, they are positioned in coastal regions exhibiting maximum deformation. Note that because InSAR exclusively resolves differential motion relative to local reference points, the resulting amplitudes are inherently smaller than the absolute GNSS displacements.

Deformation modeling and inversion

Considering the large gap in deformation measurements between islands (which results in large geophysical null space) and large uncertainty of fault/magma reservoir geometries, we forgo inverting deformation with distributed dislocations. Instead, we identify deformation sources using principles of elasticity and then proceed to perform inversions with a minimum number of sources explaining the observed deformation.

We model magma reservoirs as spheroidal cavities embedded in an elastic medium, subjected to spatially uniform, outward-normal traction changes at the cavity boundary (*66*, *67*). We model faults using rectangular dislocation elements in 3D (*68*), capable of having uniform opening, dip slip, and strike slip within each element. All modeling assumes homogeneous, isotropic, elastic half space and assumes that the free surface of the half space is at the sea level. Given that all deformation sources were found to be at least 5 km below the surface, accounting for topography and bathymetry would have had a negligible influence on inversion results.

When modeling InSAR LOS displacements, we treat each island as an independent tiltmeter with its own local reference (table S5). The reasoning is that deformation-free reference does not exist on these islands and interferometric phase (and the derived deformation) connectivity is lost between islands. Because these islands host few GNSS stations and vertical GNSS components are inherently noisy, we do not tie InSAR to absolute GNSS motion. Instead, we use relative displacement fields by referencing model predictions to the same local stationary points used in the InSAR processing. This allows us to fit the local deformation gradient directly, without introducing arbitrary constant offsets during the inversion [equation 3 of (*69*)].

We first performed a Bayesian inversion with InSAR-derived LOS displacement maps and GNSS displacements over Santorini (1 July 2024 to 26 January 2025). We model the deformation source as a magma reservoir and sampling the posterior parameter space using Markov Chain Monte Carlo with the Metropolis-Hastings criterion. Details of the inversion procedure are described in section 3 of (*70*). We use uninformative priors (uniform prior distributions within given bounds) for the inversion (table S1). The inversion ends with 5×10^6 samples collected in the posterior distribution, and the first 5×10^5 samples are discarded as burn-in. We do not weigh the residuals using a data covariance matrix derived from spatial noise, such as done in (*70*): We found that semivariograms of residuals tend to produce the largest covariance where data misfit is high. Thus, weighing residuals with the semivariogram-derived spatial covariance matrix tends to degrade data fitting where the data fit was already poor.

We then performed a nonlinear inversion with InSAR-derived LOS displacement maps and GNSS displacements over all five islands (Santorini, Anydros, Ios, Anafi, and Amorgos) for the period between 27 January 2025 and 24 February 2025. We use the MATLAB function "patternsearch" to invert for the geometry and location of the Kolumbo reservoir, the dike, and the fault (parameter bounds in tables S2 and S3). The inversion ran until the residual ceased to decrease.

For both inversions, we use InSAR LOS uncertainties propagated from the time-series fitting at each pixel to the downsampled areas to weigh the residuals (LOS displacements are spatially downsampled to 400 m–by–400 m areas, except on Anydros, where displacements are downsampled to 100 m–by–100 m areas).

Perturbation tractions and Coulomb stress

We compute the perturbation normal traction

$$\Delta T_n = \Delta\sigma_{ji} n_j n_i \quad (5)$$

due to volume increase of the Santorini reservoir on the dike plane. Here, indicial notation is used and $i, j = x, y$ or z. $\Delta\sigma_{ji}$ is the perturbation stress tensor. n_j is the unit normal vector of either side of the dike surface pointing into the surrounding crust, such that positive ΔT_n indicates that the dike-normal stress becomes more tensile.

We also compute the changes in Coulomb stress on the fault due to opening on the dike. Specifically, we compute the scalar Coulomb failure stress, ΔCFS, in the dominant direction of slip (in this case, dip-slip) on the fault

$$\Delta\mathrm{CFS} = \Delta\sigma_{ji} n_j t_i + f\,\Delta\sigma_{ji} n_j n_i \quad (6)$$

where f is the coefficient of friction, here assumed to be 0.7. t_i is the unit vector in the dominant direction of displacement, defined on the same side of the fault as n_i. All stress modeling assumes a homogeneous and isotropic half space with shear modulus of 20 GPa and Poisson's ratio of 1/4.

Estimating time-dependent aseismic and seismic moment

We compute the time-dependent scalar aseismic moment $M_0^{\mathrm{aseismic}}(t)$ for the dike and the fault for the period spanning 27 January 2025 and 24 February 2025. Spatially uniform elastic properties (shear modulus, $\mu = 25$ GPa, and Poisson's ratio, $\nu = 1/4$) and fault slip or opening are assumed.

First, we compute the cumulative aseismic moment for the fault and the dike using

$$M_0^{\mathrm{fault}} = \mu\delta_{\mathrm{slip}} A_{\mathrm{fault}} \quad (7)$$

and

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$$M_0^{\text{dike}} = \sqrt{\frac{11}{2}}\lambda\delta_{\text{opening}}A_{\text{dike}} \quad (8)$$

respectively. The factor of $\sqrt{11/2}$ arises from the definition of the scalar moment $M_0 = \left(M_{ij}M_{ij}/2\right)^{1/2}$ and the assumption of Poisson solid. A is the area for shear slip or opening. We assume that A is constant in time and is taken as that from the geodetic inversion (tables S2 and S3). The magnitude of shear slip, δ_{slip}, and the magnitude of opening, δ_{opening}, are also taken as those from the geodetic inversion (tables S2 and S3).

Next, we infer the time dependence of aseismic moment release for the dike and the fault. We use the horizontal displacement measured on GNSS station SANT on Santorini during this period to infer the time dependence of dike aseismic moment release. The reasoning is as follows. Because the Kolumbo reservoir is the primary source of magma for the dike intrusion, the time dependence of deflationary deformation at Santorini, reflecting the volume decrease of the Kolumbo reservoir, should approximately mirror the time dependence of volume increase in the dike. This is a first-order approximation, but one that nonetheless gives us a constraint on the time dependence of moment release at the dike. We normalize the displacement time series by the displacement at the end of the period to retrieve the time dependence of aseismic moment release.

Similarly, we estimate the time dependence of the aseismic moment release for the fault, using our inferred vertical displacement at the GNSS station ANYD on Anydros. The reasoning is that subsidence at Anydros is the most sensitive to faulting (see the "Dike intrusion originating from the Kolumbo volcano" section).

The GNSS station ANYD has incomplete coverage in the period spanning 27 January 2025 and 24 February 2025 (*35*). To obtain the time dependence of faulting, we first solve for the InSAR vertical displacement relative to a reference point centered on Anydros, assuming that all horizontal displacement is in the direction of GNSS horizontal displacement (Eq. 4). Because there is no stationary reference point on Anydros, the vertical displacement that we obtain is akin to tilt. We normalize the tilt history by the tilt at the end of intrusion and multiply it by a scale factor that converts the GNSS vertical displacement to that of InSAR in the same period. The resulting vertical displacement time series shows that, on 24 February 2025, Anydros had subsided by 39 cm (fig. S9). We use the time dependence of this displacement time series as the time dependence of moment.

We estimate the seismic moment using earthquakes from USGS ANSS Comprehensive Earthquake Catalog in our study area from 1 January to 30 April 2025 (*71*). Most events in the catalog have only body wave magnitudes, except the bigger events, which also have moment magnitude estimates. We perform a linear regression between moment and body wave magnitude for those larger events and use that to convert body wave magnitudes to moment magnitudes for the rest of the catalog. The scalar moment is then converted from the moment magnitude using

$$M_0 = 10^{1.5M_w+9.05} \quad (9)$$

Supplementary Materials

This PDF file includes:

Supplementary Text
Tables S1 to S6
Figs. S1 to S11
References

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